

EXTERNAL INTELLIGENCE FOR IT OPERATIONS

InfraRisk Analyzer

Review infrastructure changes before they become outages.

InfraRisk Analyzer extends the Assignment 1 Change Review Board with a request-time External Intelligence Review. An Operator selects a locally stored IT Change and enters a CVE Identifier. The application retrieves technical evidence from the NVD JSON API and an exploitation/remediation signal from the CISA Known Exploited Vulnerabilities catalog.

Boundary: External evidence informs human review. It does not prove that the selected Affected System is vulnerable, change the manually assigned Risk Level, approve or block the IT Change, or create a persistent source record.

API source

NVD API 2.0 returns the English description, publication date, best CVSS metric, and record URL.

Scraped source

CISA's public KEV table contributes vendor, product, vulnerability, dates, and required action.

Unified purpose

Both sources answer what current vulnerability evidence an Operator should consider before a change window.

InfraRisk Analyzer

Home Change Review Board

IT OPERATIONS REVIEW

Review infrastructure changes before they become outages.

InfraRisk Analyzer helps Operators compare proposed IT Changes, spot risk signals, and prepare mitigations before a maintenance window.

[Review Change Risk ->](#)

01 Risk signals

Surface production impact, security exposure, broad user impact, and missing review details before an IT Change proceeds.

02 Rollback readiness

Keep recovery planning visible so Operators can judge whether a change can be reversed if the window goes poorly.

03 Review metadata

Compare Change Tickets, owners, Review Status, Approval Records, and Audit Trail events in one scannable board.

SAMPLE RISK SUMMARY

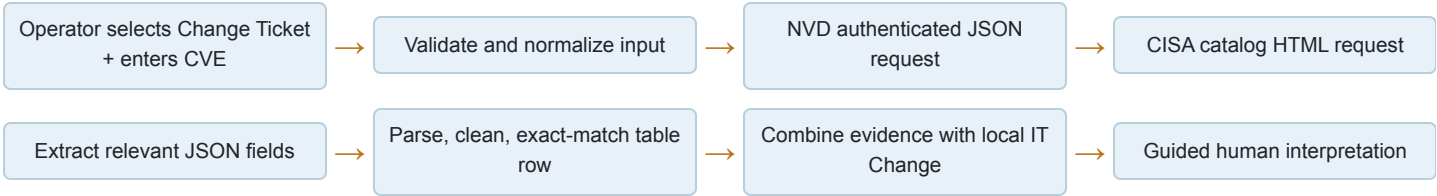
Firewall change needs security review before the window.

A High Risk firewall allow rule touches the production perimeter and depends on external source IP ranges. Mitigate by limiting scope, enabling rule logging, and confirming Security sign-off before the scheduled window.

Created by Isaac Lares for IT 401 at CSU Channel Islands. Demonstration data, not a production change-management or approval system.

Figure 1. Homepage preserves the A1 purpose and directs the Operator to Change Risk review.

One review, two external evidence boundaries



Source and environment contract

Source	Endpoint / fields	Purpose and boundary
NVD API 2.0	<code>https://services.nvd.nist.gov/rest/json/cves/2.0</code> English description, published, CVSS, NVD URL	Technical vulnerability record; uses <code>NVD_API_KEY</code> , a normalized <code>cveId</code> , User-Agent, and a finite timeout.
CISA KEV catalog	<code>https://www.cisa.gov/known-exploited-vulnerabilities-catalog</code> Vendor/project, product, name, dates, required action	Public webpage scrape; exact CVE match only. Fifteen-minute process-local cache reduces repeat requests.

Environment variables: `NVD_API_KEY` is required for NVD; `NVD_API_URL` , `CISA_KEV_URL` , and `SECRET_KEY` are optional overrides/configuration. A placeholder-only `.env.example` is committed; real `.env` is ignored.

Information model

Local IT Change

title, Change Type, Risk Level, Affected System, window, Risk Signals, Mitigation, Rollback Plan, Change Owner, Change Ticket, Review Status, Approval Records, Audit Trail.

Request-time evidence

NVD: CVE Identifier, description, published, CVSS, source URL, retrieval time.

CISA: CVE Identifier, vendor/project, product, vulnerability name, dates, required action, source URL, retrieval time.

External Intelligence Review

The selected Change Ticket remains visible beside both source panels. Guided questions separate source facts from Operator interpretation and preserve the local manually assigned Risk Level.

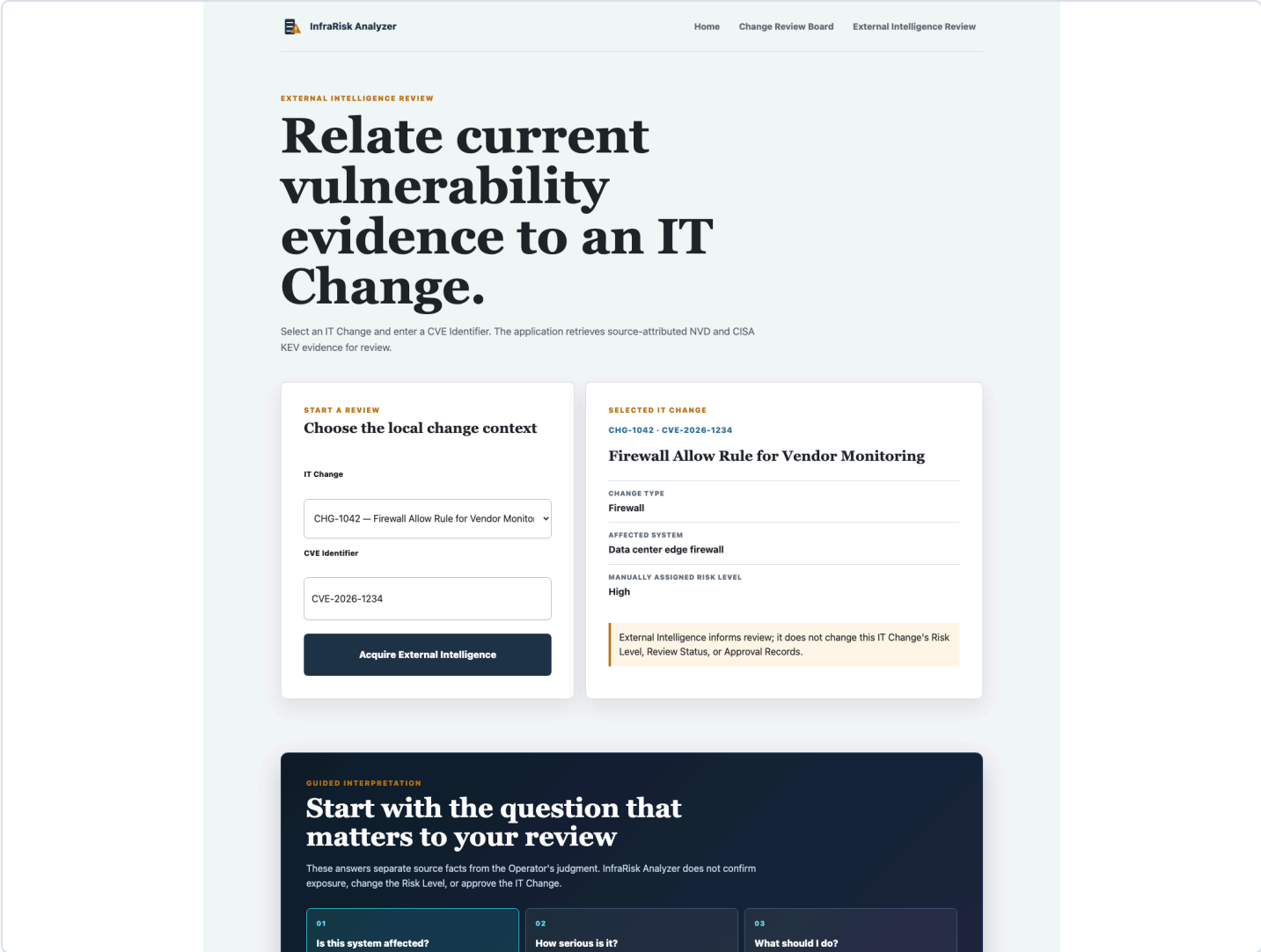


Figure 2. Combined desktop state: selected IT Change, NVD API evidence, CISA KEV catalog evidence, source details, and guided interpretation. Labels make API, scraped, and integrated requirements explicit.

Failure-handling matrix

Condition	Behavior	Test evidence
Invalid CVE or unknown Change Ticket	Rejects before acquisition with a plain-language 400/404 page; no outbound request for invalid CVE.	Route parametrized tests.
Missing NVD key	503 configuration message; NVD request is not called.	<code>test...missing_key...</code>
NVD failure, rate limit, network, malformed JSON	502 source-unavailable message, or 429 retry-later message; no traceback.	HTTP, timeout, and JSON tests.
Empty NVD result	404 "No NVD record found" message.	Empty-result route test.
Changed/missing CISA markup or network failure	503 partial state preserves NVD evidence and says CISA listing status cannot be determined.	Scraper malformed-row/header and partial-result tests.
CISA exact-match miss	Displays "not listed in this catalog snapshot"; never claims safe or non-exposed.	No-match route test.

Verification record

- Automated suite: **41 passed** with mocked sources; no live network or real credential required.
- Production routes verified in the checked-in screenshots: homepage, Change Review Board, External Intelligence Review, source details, guided questions, and narrow responsive layout.
- Reduced-motion behavior is implemented in `static/style.css`; JavaScript enhancement keeps all answers available without scripting.
- Repository scan: no tracked API key, token, password, local environment file, or secret-bearing config; only placeholder `.env.example`.
- Manual authenticated NVD live-source verification is intentionally not claimed in this artifact because this environment has no real NVD credential. It is the remaining operator-only release step after configuring a private local `.env`.

Ethics, limitations, and Assignment 3

Ethical considerations

- Attributes NVD and CISA and links to their source records.
- Uses the documented NVD endpoint, API key header, descriptive User-Agent, finite timeout, and respectful request frequency.
- Scrapes only public CISA table fields needed for review; respects source terms, licensing, robots guidance, and layout changes.
- Does not collect personal information or infer compromise from a CVE match.

Known limitations

- Evidence is request-time and process-local CISA caching is lost on restart.
- External schemas, availability, rate limits, and HTML layout can change.
- A CVE does not establish that the selected system uses the affected product/version.
- Demonstration data is not a production approval system.

Assignment 3 direction

Add persistent IT Changes, Risk Findings, Approval Records, Audit Trail events, and timestamped External Intelligence Snapshots. A snapshot will retain the normalized NVD/CISA evidence, source URLs, retrieval timestamps, selected Change Ticket, and Operator review context without turning external evidence into automatic approval or blocking logic.

Release checklist

The repository contains the A1 PDF unchanged plus this versioned A2 PDF, updated README sections, focused service and route tests, source fixtures, and checked-in browser screenshots. The final local steps are `.venv/bin/python -m pytest -q`, a private-key live NVD check, and a final secret scan before submission.